	Variant	**Macro-F1** (3-way) mean	Δ	**MCC** (binary: E vs. ¬E) mean	Δ	W/L
	COVER-PL (full)	**92.23**±0.49	–	**90.54**±0.70	–	–
Know.	w/o lexical relations (r_{ij}, o_i)	91.87±0.41	−0.36	89.50±0.62	−1.04*	1/9
Know.	w/o graph layers (c)	91.75±1.00	−0.48	89.22±1.25	−1.32*	2/8
Head	w/o conflict branch (c_k, noisy-OR)	91.45±0.99	−0.79*	89.42±1.24	−1.11**†	0/10
Head	w/o coverage branch (s_k, weakest link)	92.04±0.38	−0.20	90.16±0.76	−0.38	4/6
Head	logical head → MLP head	91.71±1.34	−0.52	89.81±1.57	−0.73	4/6
	BERTimbau [CLS] baseline (`pair_cls`)	91.76±0.60	−0.47	88.99±0.92	−1.55**	0/10

InferBR test set (1,705 pairs), BERTimbau-base, same fixed hyperparameters for every row, 10 paired seeds; mean ± s.d. in %. Δ: mean paired difference to the full model (percentage points). W/L: seeds where the variant wins/loses against the full model. Two-sided Wilcoxon signed-rank test: $^{*}p < 0.05$, $^{**}p < 0.01$; †significant after Holm correction over the five ablations. *Know.*: knowledge injection; *Head*: coverage/conflict head and logical aggregation (panels (d)–(e)).